

Exponential by Design: Why a Million Classical Cores Cannot Match a Hundred Good Qubits

HPC vs. Quantum Computers — From the Roots of Parallelism to the Exascale–NISQ Standoff

Esteban Hernández*

CyberColombia, Bogotá, Colombia

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Abstract

Classical computing has spent five decades extracting every available level of parallelism: instruction-level parallelism (ILP), deep cache hierarchies, vector units, multicore processors, multi-socket nodes, low-latency interconnects, and clusters of thousands of nodes culminating in verified exascale systems exceeding 2×10^{18} floating-point operations per second. A natural question arises: if we can already execute millions of tasks simultaneously, does the premise that quantum computing is fundamentally more powerful still hold? This paper argues that it does, but for reasons frequently misunderstood. Classical parallelism yields at most a *linear* speedup in physical resources, whereas an n -qubit quantum register inhabits a Hilbert space of dimension 2^n , growing *exponentially* with linear hardware. Crucially, this advantage is not “trying all answers at once”: it is realized only through quantum interference, and only for problems with exploitable structure. We trace both paradigms from their theoretical roots (Flynn, Amdahl, Gustafson; Feynman, Deutsch, Shor, Grover), formalize the comparison through complexity theory (P, BPP, BQP), survey the principal qubit implementation technologies and their scaling prospects, and confront theory with the 2026 reality: exascale supercomputers with millions of cores versus noisy intermediate-scale quantum (NISQ) devices with hundreds of physical qubits. We conclude that the future is not replacement but hybridization: quantum processing units as accelerators within classical HPC infrastructures.

Keywords: high-performance computing, quantum computing, parallelism, computational complexity, NISQ, exascale, qubit technologies.

1 Introduction

The history of computing performance is, to a first approximation, the history of parallelism. When frequency scaling collapsed against the power wall in the mid-2000s, the industry’s answer was to multiply: more pipelines, more cores, more sockets, more nodes. Today the TOP500 list is headed by machines sustaining more than two exaflops on the High-Performance Linpack (HPL) benchmark, aggregating millions of cores across thousands of nodes connected by sub-microsecond interconnects.

Against this backdrop, the standard claim that “quantum computers are exponentially more powerful” deserves scrutiny. If a classical supercomputer already executes 10^7 hardware threads concurrently, in what precise sense can a device with a few hundred fragile qubits be more powerful? This paper answers that question rigorously. Section 2 reconstructs the full classical

*Corresponding author: esteban.hernandez@cybercolombia.org

parallelism stack and its formal limits. Section 3 presents the roots of quantum parallelism and dismantles the popular but incorrect “massive parallel evaluation” narrative. Section 4 formalizes the comparison via computational complexity. Section 5 surveys qubit implementation technologies and their near-term scaling prospects. Section 6 confronts exascale HPC with the NISQ era. Section 7 argues for the hybrid HPC+QC model, and Section 8 concludes.

2 The Roots and Limits of Classical Parallelism

2.1 The parallelism stack

Modern classical performance is the product of parallelism exploited simultaneously at every architectural level:

Instruction-level parallelism (ILP). Pipelining, superscalar issue, out-of-order execution, branch prediction, and speculative execution allow a single core to retire multiple instructions per cycle. ILP is bounded by data dependencies and control hazards; empirical studies place sustainable IPC in the low single digits for general-purpose code.

Data-level parallelism (DLP). SIMD vector units (SSE/AVX-512, SVE, GPU warps) apply one instruction to wide registers or thousands of lanes. GPUs generalize DLP into SIMT execution with tens of thousands of concurrent threads per device.

Memory hierarchy. Multi-level caches (L1/L2/L3), prefetchers, and NUMA-aware memory controllers exist to feed the parallel functional units; effective parallelism is frequently bounded not by compute but by memory bandwidth (the “memory wall” and the roofline model).

Thread- and task-level parallelism. Multicore and simultaneous multithreading within a socket; multiple sockets per node with coherent shared memory (ccNUMA).

Node- and cluster-level parallelism. Distributed-memory clusters programmed with MPI, connected by low-latency fabrics (InfiniBand, Slingshot) achieving microsecond-scale latencies and hundreds of Gb/s per port; at the extreme, TOP500-class systems with thousands of nodes and millions of cores.

2.2 Formal limits: Amdahl and Gustafson

Let f be the serial fraction of a program. Amdahl’s law bounds the speedup with P processors:

$$S(P) = \frac{1}{f + \frac{1-f}{P}} \xrightarrow{P \rightarrow \infty} \frac{1}{f}. \quad (1)$$

Gustafson’s law rehabilitates scaling for problems whose size grows with P ($S(P) = P - f(P - 1)$), which is precisely how weak-scaling HPC workloads are designed. Yet both laws share an inescapable structural property: *the speedup is at most linear in P* . Doubling capability requires doubling hardware, energy, and cost. In asymptotic terms, a parallel machine with P processors can reduce the time of a T -step computation to at best T/P ; it cannot change the *complexity class* of a problem. A parallel computer is polynomially equivalent to a sequential one.

3 The Roots of Quantum Parallelism

3.1 Feynman’s motivation and Deutsch’s formalization

Quantum computing was born from a limitation of classical parallelism. Feynman (1982) observed that simulating a quantum system of n particles requires classical resources growing exponentially in n , and proposed that only a computer operating under quantum rules could simulate nature efficiently. Deutsch (1985) formalized the universal quantum Turing machine and the quantum circuit model.

3.2 State space: exponential by design

A register of n qubits is described by a unit vector in a Hilbert space of dimension 2^n :

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \sum_x |\alpha_x|^2 = 1, \quad \alpha_x \in \mathbb{C}. \quad (2)$$

Adding *one* qubit *doubles* the dimension. A single unitary gate acts linearly on all 2^n amplitudes simultaneously. This is the precise sense of “quantum parallelism”: linear growth in hardware yields exponential growth in the descriptive state space. No classical architecture shares this property; a classical machine with P processors has a state space that grows *polynomially* in useful capacity with P .

3.3 Why the popular narrative is wrong

It is tempting—and widespread—to conclude that a quantum computer “evaluates 2^n solutions in parallel and is therefore 2^n times faster.” This is false, for three reasons:

1. **Measurement.** Reading the register collapses the state and returns a *single* outcome x with probability $|\alpha_x|^2$. The 2^n amplitudes are not individually accessible.
2. **The Holevo bound.** At most n classical bits of information can be extracted from n qubits.
3. **No-cloning.** Quantum information cannot be copied to circumvent measurement.

The true source of quantum advantage is *interference*: algorithms are choreographed so that amplitudes of wrong answers cancel destructively while amplitudes of correct answers reinforce constructively before measurement. This requires exploitable mathematical structure in the problem. Where such structure exists (period finding in Shor’s algorithm via the quantum Fourier transform), the speedup can be exponential; where it does not (unstructured search), the best possible speedup is quadratic (Grover), and this bound is provably optimal.

4 Formal Comparison: Complexity Theory

4.1 Classes and separations

Let P, BPP, and BQP denote the classes of problems solvable in polynomial time deterministically, probabilistically, and by a quantum computer with bounded error, respectively. It is known that $P \subseteq BPP \subseteq BQP \subseteq PSPACE$. The widely believed—but unproven—conjecture $BPP \subsetneq BQP$ is the formal statement of quantum advantage. Its strongest evidence is Shor’s algorithm: integer factorization lies in BQP with cost $O((\log N)^3)$, whereas the best classical algorithm, the general number field sieve, is sub-exponential:

$$L_N \left[\frac{1}{3}, \sqrt[3]{64/9} \right] = \exp \left(\left(\sqrt[3]{64/9} + o(1) \right) (\ln N)^{1/3} (\ln \ln N)^{2/3} \right). \quad (3)$$

No amount of classical parallelism changes this picture: distributing a sub-exponential algorithm over P processors divides the constant, not the exponent.

4.2 The extended Church–Turing thesis

The deeper claim at stake is the *extended Church–Turing thesis*: that any physically realizable model of computation can be simulated by a probabilistic Turing machine with polynomial overhead. Quantum computing is the only credible physical challenge to this thesis. Classical parallelism, by contrast, is fully contained within it.

4.3 Grover and the limits of generic speedup

Grover’s algorithm finds a marked item among N with $O(\sqrt{N})$ oracle queries, versus $\Theta(N)$ classically, and the quadratic bound is optimal (Bennett–Bernstein–Brassard–Vazirani). This is a critical honesty point for any comparison: for generic search and for most dense linear algebra, CFD, and molecular dynamics workloads that dominate HPC centers, no exponential quantum speedup is known, and classical exascale systems will remain superior for the foreseeable future.

4.4 The cost of classical simulation of quantum systems

The converse comparison quantifies the asymmetry. Storing the state vector of n qubits in double precision requires 2^{n+4} bytes: 45 qubits $\approx$ 0.5 PB; 50 qubits $\approx$ 16 PB; 53 qubits exceed the aggregate memory of any machine on the TOP500. Tensor-network contraction methods have repeatedly pushed this frontier—notably re-simulating the 2019 “quantum supremacy” circuits far faster than originally claimed—but their cost still grows exponentially with circuit entanglement. The 2019–2023 supremacy-vs-simulation exchange between Google and classical simulation teams is itself an elegant case study of this paper’s central tension.

5 Qubit Implementation Technologies and Scaling Prospects

The theoretical advantage of Section 4 is only realizable if qubits can be manufactured, controlled, and error-corrected at scale. This chapter surveys the principal technologies and evaluates, on current evidence, which offer the most credible near-term scaling path.

5.1 Superconducting transmon qubits

Josephson-junction circuits operated at ~ 15 mK in dilution refrigerators. **Strengths:** fast gates (10–100 ns), lithographic fabrication leveraging semiconductor processes, the most mature ecosystem (IBM, Google, Rigetti); IBM’s Heron generation introduces tunable couplers to suppress crosstalk, and its public roadmap targets modular fault tolerance (Kookaburra 2026, Cockatoo 2027, Starling $\sim$ 2029 with $\sim$ 200 logical qubits and 10^8 operations). **Weaknesses:** short coherence times (tens–hundreds of μ s), two-qubit gate fidelities around 99–99.9%, only nearest-neighbor connectivity, and massive cryogenic wiring overhead per qubit. **Scaling outlook:** strong in the near term via chip modularity and qLDPC error-correcting codes that reduce the physical-to-logical qubit overhead by roughly an order of magnitude relative to surface codes.

5.2 Trapped ions

Atomic ions (Yb^+ , Ca^+) confined in electromagnetic traps, manipulated by lasers (IonQ, Quantinuum). **Strengths:** the highest gate fidelities demonstrated ($>99.8\%$ two-qubit), long coherence (seconds–minutes), all-to-all connectivity within a trap, identical qubits by nature. **Weaknesses:** slow gates (μ s–ms), and traps saturate at tens of ions; scaling requires QCCD shuttling architectures or photonic interconnects between traps, both engineering-intensive. **Scaling outlook:** excellent quality but slower growth in qubit count; the leading platform for early logical-qubit demonstrations, plausible mid-term path via modular photonic networking.

5.3 Neutral atoms

Arrays of atoms (Rb, Sr) held in optical tweezers, entangled via Rydberg states (Pasqal, QuEra, Atom Computing). **Strengths:** demonstrated arrays beyond 1,000 physical atoms, reconfigurable geometry (2D/3D), room-temperature vacuum apparatus rather than dilution refrigeration, natural fit for analog quantum simulation. **Weaknesses:** gate fidelities historically below ions and transmons (improving rapidly), atom loss, slow cycle rates. **Scaling outlook:** arguably the fastest raw-qubit-count scaling demonstrated to date; recent logical-qubit experiments (transversal gates on dozens of logical qubits) have moved this platform, unexpectedly, to the front of the error-correction race. A leading candidate if fidelities keep improving.

5.4 Photonic qubits

Information encoded in photons processed through linear optics and measurement-based computation (PsiQuantum, Xanadu). **Strengths:** room-temperature operation (except detectors), natural networking over optical fiber, leverages silicon-photonics foundries. **Weaknesses:** photons do not interact directly—two-qubit gates are probabilistic and demand enormous multiplexing overhead; photon loss is the dominant error. **Scaling outlook:** a high-risk, high-reward “all-or-nothing” bet on manufacturing scale; if foundry-based fusion networks succeed, million-qubit systems are conceivable, but intermediate milestones are scarce.

5.5 Silicon spin qubits

Electron or nuclear spins in silicon quantum dots or donor atoms (Intel, Diraq, UNSW). **Strengths:** qubit footprint of nanometers, direct compatibility with CMOS manufacturing—the only technology that could ride the existing semiconductor industry; recent results show nuclear-spin coherence beyond 30 s and error rates below 1%, with entanglement demonstrated between donor nuclei at 20 nm separation. **Weaknesses:** currently small device counts (tens of qubits), variability between dots, control electronics integration at millikelvin. **Scaling outlook:** potentially the best long-term scaling story (billions of transistors already fit on a die), but the furthest from demonstrating intermediate-scale systems.

5.6 Topological qubits

Qubits encoded in non-Abelian anyons (Majorana zero modes), pursued by Microsoft. **Strengths:** error protection built into the physics—topological encoding would suppress local noise intrinsically, slashing error-correction overhead. **Weaknesses:** the existence and control of the underlying quasiparticles remained contested for years; devices are at the few-qubit validation stage. **Scaling outlook:** transformative if it works; the highest scientific risk of all platforms.

5.7 Comparative assessment

Technology	2Q fidelity	Gate speed	Demonstrated scale	Connectivity	Near-term scaling
Superconducting	99–99.9%	10–100 ns	~100–1,000+ phys.	nearest-neighbor	High
Trapped ions	>99.8%	μ s–ms	tens per trap	all-to-all	Medium
Neutral atoms	99–99.5%	μ s	>1,000 atoms	reconfigurable	High
Photonic	n/a (fusion)	ns	components	network-native	Uncertain
Silicon spin	~99%	ns– μ s	tens	nearest-neighbor	Long-term high
Topological	—	—	validation	—	Speculative

Table 1: Qualitative comparison of qubit platforms (representative 2026 values).

On present evidence, **superconducting circuits and neutral atoms offer the most credible near-term scaling paths**—the former through industrial roadmaps, modular chip coupling, and efficient qLDPC codes; the latter through massive, reconfigurable atom arrays and rapid progress in logical-qubit operations. Trapped ions lead on fidelity and will likely host the highest-quality early logical qubits. Silicon spins are the strongest *long-term* bet because they inherit the entire CMOS manufacturing base. The decisive metric in all cases is not physical qubit count but the ratio of physical to logical qubits and the logical error rate per operation.

6 The 2026 Reality: Exascale vs. NISQ

6.1 Classical: verified exascale

The June 2026 TOP500 list is headed by LineShine (National Supercomputing Center, Shenzhen) at 2.198 exaflops HPL—notably a CPU-only architecture—followed by El Capitan (LLNL) at 1.809 exaflops, with Frontier and Aurora also above one exaflop. These systems aggregate on the order of 10^7 cores, consume 30–42 MW, and rest on fifty years of mature software: MPI, OpenMP, CUDA/HIP, resilient parallel file systems, and checkpoint-restart fault tolerance.

6.2 Quantum: the NISQ era

Quantum hardware in 2026 remains in the regime Preskill named NISQ: noisy devices of roughly 50 to $\sim 1,000$ physical qubits, without full error correction, where decoherence limits executable circuit depth to thousands of gates. Physical qubits are not logical qubits: surface-code error correction demands on the order of 10^2 – 10^3 physical qubits per logical qubit at current error rates, which is why vendor roadmaps (e.g., IBM Starling, ~ 2029 , ~ 200 logical qubits and 10^8 quantum operations) treat the fault-tolerant machine as an end-of-decade goal rather than a present reality.

6.3 The asymmetry of scaling difficulties

The comparison exposes a striking asymmetry. Classical HPC scales easily in hardware but is asymptotically capped: its speedup is linear in resources and it cannot leave its complexity class. Quantum computing is asymptotically unbounded on structured problems but scales with enormous difficulty in hardware: adding qubits adds noise, crosstalk, wiring, and cryogenic load, and the error-correction tax must be paid before any asymptotic advantage materializes. Millions of robust cores face hundreds of fragile qubits; the race is between engineering maturity and mathematical destiny.

7 Convergence: The Hybrid HPC+QC Model

The realistic future is not replacement but integration, mirroring how GPUs entered the node two decades ago. Quantum processing units (QPUs) are already being co-located with supercomputers (e.g., quantum systems installed at Jülich and national laboratories) and exposed through hybrid runtimes in which the classical machine performs pre-processing, variational parameter optimization, error mitigation, and post-processing, while the QPU executes the classically intractable kernel: quantum simulation of molecules and materials, period finding, and, prospectively, quantum linear-algebra subroutines. The end state of this paper’s titular confrontation is thus an architecture diagram, not a victory: the exascale cluster with a cryostat on the machine-room floor.

8 Conclusions

(1) Classical parallelism, from ILP to exascale, delivers at most linear speedup in physical resources and cannot alter a problem’s complexity class. (2) Quantum parallelism is not parallel evaluation but exponential state space plus interference; it yields super-polynomial advantage only for structured problems, exponential in the best cases (factoring, quantum simulation) and merely quadratic for unstructured search. (3) The premise that quantum computing is potentially more powerful therefore survives exascale intact—indeed exascale itself, unable to simulate more than ~ 50 qubits exactly, is the cleanest empirical witness of the gap. (4) Among implementation technologies, superconducting circuits and neutral atoms currently present the most credible near-term scaling paths, trapped ions the highest fidelity, and silicon spins the strongest long-term manufacturing story; the governing metric is logical, not physical, qubits. (5) The practical trajectory is hybrid: QPUs as accelerators inside classical HPC, each paradigm doing what the other provably cannot.

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